Day 2

General set-up code

% Init.

Parameters; Define ALL #s as parameters.

Structures; definitions?? functions?

% Main

for nsteps

% update % For large problem direct → write in comments → code

% Skeleton → slowly flesh out

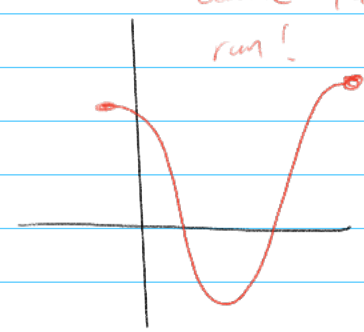
% viz / save data % viz helps to debug!

End

% save data

code if statement could fail to run!

Go over code. what happens when



<u>Def</u> $\{x_n\}_{n=1}^{\infty} \to x^*$. The <u>absolute error</u> b/w x^* and the approx. value x_k is the error given by $|x_k - x^*|$.

<u>Def</u> If $x^* \neq 0$ <u>relative error</u> is $\dfrac{|x_k - x^*|}{|x^*|}$

<u>Thm</u> If $f \in C[a,b]$ and $f(a)f(b) < 0$. The bisection alg generates $\{[a_n, b_n]\}_{n=1}^{\infty}$ and the associated midpoints

$$x_n = \frac{a_n + b_n}{2}$$

The sequences $\{a_n\}, \{b_n\}, \{c_n\}$ all converge to x^* of the eq.

$f(x^*) = 0$ and the abs. error

$$|x_k - x^*| \le \frac{1}{2}(b_k - a_k) = 2^{-(k+1)}(b-a)$$

WTS $a_n, b_n, x_n \longrightarrow x^*$, $\exists x^*$, st. $f(x^*) = 0$

$$|x_n - x^*| \le \frac{1}{2^{n+1}}|a-b|$$

Pf: Note $\{a_n\}$ is monotonically increasing and has upper bnd b.
$\Rightarrow a_n$ converges i.e. $A = \lim_{n\to\infty} a_n$

Similarly $\{b_n\} \downarrow$ mon and bnd below by $a \Rightarrow b_n$ converges.
i.e. $B = \lim_{n\to\infty} b_n$

~~$a_n \le x_n \le b_n \Rightarrow x_n$ is bounded~~

$$|A-B| \le \lim_{n\to\infty} |a_n - b_n| = \lim_{n\to\infty} \frac{1}{2}|a_{n-1} - b_{n-1}| = \cdots \lim_{n\to\infty} \frac{1}{2^n}|a_0 - b_0|$$
$$= \lim_{n\to\infty} \frac{1}{2^n}|a-b| = 0$$

$\Rightarrow A = B$ let $x^* = A = B$

end of 1st claim

Since $a_n \le x_n \le b_n$ $\quad a_n, b_n \longrightarrow x^* \Rightarrow x_n \to x^*$ (by squeeze thm.)

[By bisection alg.] $f(a)f(b) < 0$ + f is continuous + $f(a_n)f(b_n) \le 0$ $\forall n$

$\Rightarrow \lim_{n\to\infty} f(a_n)f(b_n) = f^2(x^*) \le 0$ (by assumption)

end of 2nd claim $\Rightarrow f(x^*) = 0$

Finally since $x^* \in \bigcap_{n=0}^{\infty} [a_n, b_n]$ and $x_n \in [a_n, b_n] \cap [a_{n+1}, b_{n+1}]$

$$\Rightarrow |x_n - x^*| \le |b_{n+1} - a_{n+1}| = \frac{1}{2}|b_n - a_n| = \cdots \frac{1}{2^{n+1}}|b_0 - a_0|$$
$$= \frac{1}{2^{n+1}}|b-a| \quad \#$$

Def: The # x^* is a fixed point for a given function g if $g(x^*) = x^*$. The problem of finding x^* s.t. $g(x^*) = x^*$ is called a fixed-point problem

ex: $g(x) = x^2 - 2$ $\quad g(-1) = -1$, $g(2) = 2$ $\quad -1, 2$ are two-fixed points of g.

Remark: Root problems are equivalent to fixed point problem

$$f(x) = 0 \iff g(x) = x + \alpha f(x)$$

Thm: ① If $g(x) \in C[a,b]$ and $g(x) \in [a,b]$ for any $x \in [a,b]$ (g is a contraction mapping on $[a,b]$), then $\exists x^* \in [a,b]$ s.t. $g(x^*) = x^*$

② If, in addition, $g \in C'(a,b)$ and $\exists k \in (0,1)$ s.t. $|g'(x)| \le k < 1$ $\forall x \in (a,b)$ then the fixed point is unique.

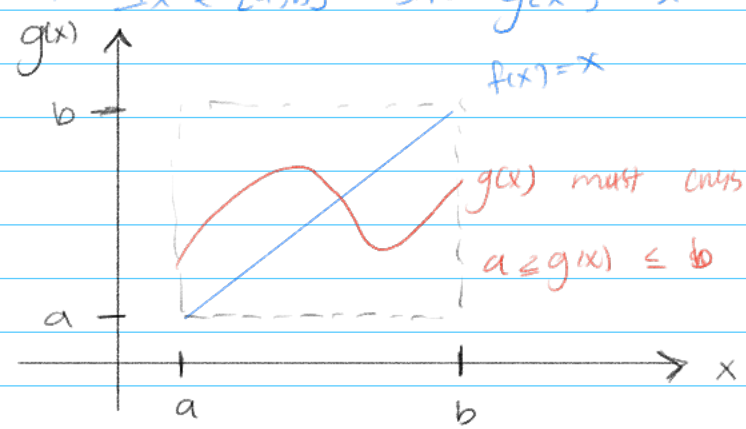
Intuition ① $g(x) \in C[a,b]$ $\quad g(x) \in [a,b]$ $\quad g: [a,b] \to g([a,b])$

i.e

where $g([a,b]) \subseteq [a,b]$ $\quad Im(g) \subseteq$ Domain

g - contraction mapping on $[a,b]$

$\Rightarrow \exists x^* \in [a,b]$ s.t. $g(x^*) = x^*$



$g(x)$ must cross $f(x) = x$ since $a \le g(x) \le b$

$g(x)$ always changes "mildly"

② Additionally $g \in C'[a,b]$ & $\exists k \in (0,1)$ s.t. $|g'(x)| \le k < 1$ $\forall x \in (a,b)$

$\Rightarrow \exists! x^* \in [a,b]$, i.e. unique